

Neural network algorithm for under-sampled wavefront reconstruction: mathematical analysis and implementation

ZHIYUN ZHANG,^{1,2,3} RUIYAN JIN,^{1,2} FANGFANG CHAI,^{1,2,4}
ZHIHAO LEI,^{1,2,4} LINXIONG WEN,^{1,2} SHUAI WANG,^{1,2}
AND PING YANG^{1,2,*}

¹National Laboratory on Adaptive Optics, Chengdu 610209, China

²Institute of Optics and Electronics, Chinese Academy of Sciences, Chengdu, Sichuan 610209, China

³School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China

⁴University of Chinese Academy of Sciences, Beijing 100049, China

*pingyang2516@163.com

Abstract: The Shack-Hartmann wavefront sensor (SHWFS) is critical in adaptive optics (AO) for measuring wavefronts via centroid shifts in sub-apertures. Under extreme conditions like strong turbulence or long-distance transmission, wavefront information degrades significantly, leading to undersampled slope data and severely reduced reconstruction accuracy. Conventional algorithms struggle in these scenarios, and existing neural network approaches are not sufficiently advanced. To address this challenge, we propose a mathematically interpretable neural network-based wavefront reconstruction algorithm designed to mitigate the impact of slope loss. Experimental results demonstrate that our algorithm achieves what is believed to be unprecedented fidelity in full-aperture aberration reconstruction with up to 70% wavefront undersampling, representing a precision improvement of approximately 89.3% compared to modal methods. Moreover, the algorithm can be fully trained using simulation data alone, eliminating the need for real data acquisition and significantly enhancing practical applicability.

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1. Introduction

In modern optical systems, the SHWFS is extensively utilized across a multitude of disciplines including AO [1–3], astronomy [4,5], ophthalmology [6,7], optical calibration, and commercial optical testing, playing an indispensable role [8]. As a sensor based on slope measurements, the SHWFS employs a microlens array coupled with a CCD photodetector to segment the incident light beam for sampling and slope calculations [9,10]. This process facilitates the acquisition of average wavefront slope information within each sub-aperture, enabling the precise reconstruction of the wavefront through reconstruction algorithms [11–14].

In the realm of high-precision applications, especially in scenarios such as laser atmospheric transmission, astronomical observation, or long-distance laser communications, the interference of atmospheric turbulence and transmission distance on wavefront detection is particularly pronounced [15,16], which lead to complex dynamic distortions in the wavefront, causing significant fluctuations in the near-field beam intensity. These fluctuations can exceed the dynamic response range of the SHWFS, resulting in undersampling of the wavefront, and thus seriously affecting the AO system's ability to pre-compensate for aberrations [17].

In response to the challenges of wavefront detection under extreme conditions, both domestic and international researchers have devoted considerable efforts towards developing more effective solutions [18–20].

We et al. conducted a comprehensive review of traditional processing methods for the SHWFS under conditions of missing sub-apertures [21]. Some researchers set missing slopes to zero for a quick preliminary wavefront estimate, but this reduced reconstruction accuracy due to oversimplification. Others excluded missing slopes and used remaining data for reconstruction by redefining the mathematical model and optimizing the computational matrix. However, this approach can significantly increase computation time due to the variability in missing sub-aperture locations, making it unsuitable for real-time AO systems. Despite their distinctive features, these methods are primarily suited for restoring wavefronts with minimal missing slopes and struggle with high undersampling ratios.

The reconstruction of full-aperture wavefronts from undersampled light spots constitutes a nonlinear fitting problem in mathematical terms. With the rapid advancement of machine learning technologies in recent years, neural networks, with their deep structures and nonlinear characteristics, demonstrate exceptional capabilities in nonlinear pattern recognition and parameter fitting within high-dimensional data [22]. Indeed, neural networks are validated as one of the most powerful mathematical tools available for addressing nonlinear fitting challenges.

In the field of AO, researchers began to extensively explore the use of neural networks to address various issues, underscoring their significant potential and practical value in wavefront reconstruction and aberration compensation. Ma et al. developed a turbulence aberration reconstruction method based on intensity images using GoogLeNet [23]. Wu et al. developed a CNN-based model that used SHWFS intensity images with bad spots as input [24]. He et al. designed the SH-ResNet model, which can effectively predict wavefront Zernike coefficients from irregular sub-aperture spots [25]. Ge et al. were the first to introduce the concept of unsupervised learning into object-independent wavefront sensing [26]. Nevertheless, research on the reconstruction of full-aperture aberrations from undersampled wavefronts is still in its early stages.

To address this issue, Zhao et al. employed the classic U-Net network model for centroid prediction [27]. This model took missing x and y slopes as inputs and, through network fitting, output a complete matrix of x and y slopes, thereby effectively reconstructing the undersampled wavefront. Although this method provides a solution to the undersampling problem to a certain extent, several issues still need further investigation.

Firstly, the lack of in-depth study on the network or its relationship with the performance of slope prediction, which limits potential improvements and optimizations of the model [28,29]. Secondly, while the method demonstrates practical application value, it still relies on training with real data, thus casting doubt on its practicality. Lastly, in circumstances where significant thermal effects from laser transmission and severe atmospheric turbulence are present, the SHWFS may encounter sub-aperture under-sampling exceeding 50%. However, the practical application of this method in engineering environments may be limited due to the fact that such extreme conditions were not been considered.

In response to the challenges of full-aperture aberration reconstruction from highly undersampled wavefront information, we conducted an in-depth investigation into the internal workings of neural networks, specifically the operation of convolutional layers and nonlinear activation layers, and their relationship with the structure of input data [30]. Through an analysis of the computation for missing slopes, this research revealed how neural networks utilize existing information for data augmentation. Building on this analysis, an interpretable convolutional neural network algorithm was proposed for the reconstruction of full-aperture aberrations from undersampled wavefront information.

Further, given that this analysis highlights the neural network's capability to process slope data at varying sampling rates—a process solely dependent on the network's internal structure—it provides theoretical support for employing simulation data as a training set. This significantly reduces the complexity of training and establishes a more convenient and practical methodological

foundation for neural networks. The theory was validated through a series of multi-dimensional numerical simulations.

Building upon the preceding analysis, we developed “UWR-Net”, an innovative neural network architecture grounded in mathematical principles. By optimizing convolutional parameters in the initial fitting and feature mapping modules to better adapt to undersampled wavefront data, we significantly enhanced the network’s nonlinear expressivity and computational efficiency. Leveraging fixed parameters post-training and high-performance deployment with Nvidia TensorRT, UWR-Net achieves inference speeds exceeding 700 frames per second on an RTX 4070, thereby satisfying the real-time requirements of adaptive optics systems. Notably, we achieved the highest reported level of full-aperture aberration reconstruction to date, with an undersampling ratio of up to 70%.

In addition, an experimental platform was constructed based on simulation requirements to generate multiple sets of wavefront data under realistic conditions. The results confirmed that the network model, trained with simulation data, could achieve high-precision reconstruction of undersampled wavefronts, further validating the effectiveness and practicality of the theoretical analysis proposed in this paper.

The subsequent organization is structured as follows:

Section 2 delineates the causes and characteristics of the wavefront undersampling problem, as well as its impact on the performance of AO systems, laying the foundation for further analysis.

Section 3 introduces the mathematical reasoning and its validation for the missing slopes “activated” in neural network. Through a detailed analysis and mathematical modeling of the internal computation propagation within neural networks, we demonstrate how neural networks effectively address the wavefront slope undersampling issue through convolutional computations and nonlinear activations.

Section 4 details the improved design of the network architecture, based on the mathematical reasoning presented earlier. This section includes not only the details of the network structure design but also validates the enhanced model’s capability to reconstruct undersampled wavefronts under various intensities of turbulence through simulation.

Section 5 presents the experimental part. In the experiments, a series of random wavefronts were generated and collected using a turbulence generator. By comparing the reconstruction capabilities of the proposed method with other methods, the significant advantages of the proposed network model in full-aperture aberration reconstruction of undersampled wavefronts were further demonstrated.

Finally, a summary is made in Section 6.

2. Challenges in wavefront detection under conditions of near-field intensity fluctuations

In AO systems, the application of SHWFS not only quantifies the distortions in the wavefront but more critically, provides real-time, precise feedback for the system’s correction and optimization. This feedback serves as the cornerstone for compensating distortions within the AO system. However, in conditions where the beam experiences near-field intensity fluctuations and dynamic changes, the morphology of the subaperture spots deteriorates. This degradation leads to increased errors in slope calculations, ultimately diminishing the precision of wavefront reconstruction.

In the SHWFS, a series of finely crafted microlenses subdivides the incoming beam into numerous minute subapertures. Each subaperture’s beam, upon focusing through a microlens, forms a well-organized array of spot images on a CCD camera. By precisely measuring the displacements of these spots in two dimensions, the average slopes of the beam wavefront within each subaperture can be calculated. According to principles of geometrical optics, a definitive linear relationship exists between these centroid shifts of the subspots and the x and y directional

slopes of the wavefront [31]:

$$\frac{\partial \Phi}{\partial x}(x_c, y_c) \cong \frac{\Delta x}{L} \quad (1)$$

$$\frac{\partial \Phi}{\partial y}(x_c, y_c) \cong \frac{\Delta y}{L} \quad (2)$$

Within this framework, $\phi(x, y)$ represents the wavefront function, while Δx and Δy denote the measured displacements of the spot centroids in the x and y directions, respectively. L signifies the distance between the sampling and detection planes.

By sequentially calculating the displacements of each subaperture's spot centroid (x_c, y_c) , relative to the reference centroid (x_r, y_r) , the equation ($\Delta x = x_c - x_r$, $\Delta y = y_c - y_r$) facilitates the determination of the average slope of the wavefront within each subaperture. The centroid calculation equation for the subspots is defined as follows [32]:

$$x_c = \frac{\sum_i^n \sum_j^m x_{ij}(I_{ij} - b)}{\sum_i^n \sum_j^m (I_{ij} - b)}, y_c = \frac{\sum_i^n \sum_j^m y_{ij}(I_{ij} - b)}{\sum_i^n \sum_j^m (I_{ij} - b)} \quad (3)$$

In the equation, (x_c, y_c) represents the centroid coordinates; $x_{i,j}$, $y_{i,j}$ denotes the pixel positions; $I_{i,j}$ corresponds to the pixel values within the subaperture; n and m are the dimensions of the SHWFS subaperture window; and b is the threshold for image denoising processes.

After successfully detecting and acquiring the wavefront slope matrix with the SHWFS, wavefront reconstruction is typically performed using modal method. The core principle of this technique involves decomposing complex wavefront distortions into a series of predefined basic modes (commonly employing Zernike polynomials, among others) in a weighted combination. Each mode represents a specific form of distortion, as indicated in Eq. (4). Here, a_i denotes the coefficients for each mode, and ε represents the detection error. Modal method provides an intuitive physical framework for analyzing and reconstructing wavefront distortions, occupying a significant role in the field of AO.

$$\varphi(x, y) = a_0 + \sum_{k=1}^n a_k Z_k(x, y) + \varepsilon \quad (4)$$

The process of wavefront reconstruction using modal method depends on the precise detection of wavefront slopes by the SHWFS. In modal method, this process begins by calculating the initial phase of each order mode corresponding to the average slope at the sensor, and based on this, constructing a modal coefficient reconstruction matrix. Using the wavefront slope data obtained by the SHWFS, a linear system of equations is formed. In a SHWFS system incorporating N orders of Zernike polynomials and M effective subapertures, the computation for wavefront reconstruction can be expressed as follows [33,34]:

$$\begin{bmatrix} G_x(1) \\ G_y(1) \\ G_x(2) \\ G_y(2) \\ \dots \\ G_x(m) \\ G_y(m) \end{bmatrix}_{2m \times 1} = \begin{bmatrix} F_{x1}(1) & F_{x2}(1) & \dots & F_{xn}(1) \\ F_{y1}(1) & F_{y2}(1) & \dots & F_{yn}(1) \\ F_{x1}(2) & F_{x2}(2) & \dots & F_{xn}(2) \\ F_{y1}(2) & F_{y2}(2) & \dots & F_{yn}(2) \\ \dots & \dots & \dots & \dots \\ F_{x1}(m) & F_{x2}(m) & \dots & F_{xn}(m) \\ F_{y1}(m) & F_{y2}(m) & \dots & F_{yn}(m) \end{bmatrix}_{2m \times n} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ \dots \\ a_n \end{bmatrix}_{n \times 1} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \dots \\ \varepsilon_{2m} \end{bmatrix} \quad (5)$$

This process can be simplified into the following form, where $\mathcal{G}$ represents the slope matrix measured by the SHWFS, $\mathcal{D}$ is the modal coefficient reconstruction matrix, ε represents the

detection error, and $\mathcal{A}$ is the modal coefficient matrix:

$$G = DA + \varepsilon \quad (6)$$

Therefore, by solving the equation using the least squares method, the modal coefficient matrix $\mathcal{A}$ can be obtained, thereby calculating the wavefront phase. Although different types of aberration modes slightly vary in their physical significance, their reconstruction processes share a similar mathematical modeling structure. All depend critically on the accurate detection of wavefront slopes by the SHWFS, a key prerequisite for precise wavefront reconstruction.

However, in practical applications of AO systems, particularly under conditions of strong turbulence and long-distance beam transmission, beams in near-field regions often undergo severe intensity fluctuations. As illustrated in Fig. 1(b), due to the limited dynamic range of the sensor, intense fluctuations often lead to partial subaperture defects, such as missing or malformed spots. The distortion and deformation, or even absence, of subspots directly result in significant errors when calculating wavefront slopes for corresponding subapertures. Because of these computational errors, the effectiveness of modal method is notably compromised.

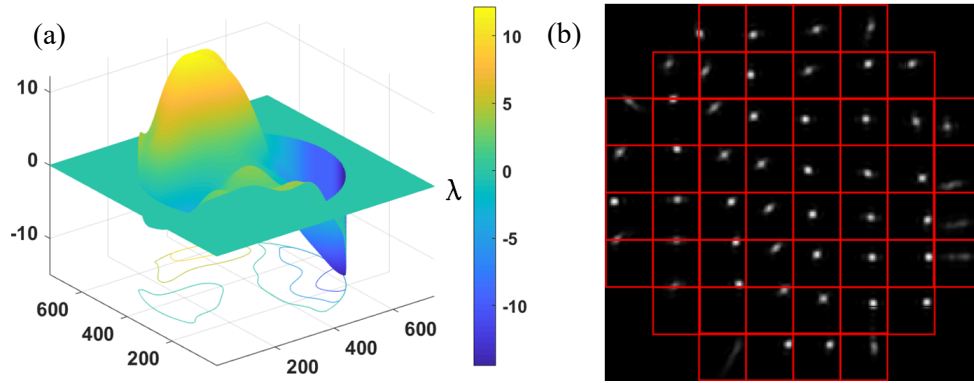


Fig. 1. Typical wavefront and corresponding subspot array images from a SHWFS. (a) Wavefront of a laser beam affected by strong turbulence. (b) Sub-aperture spot images with undersampling in certain regions.

The issues described above can lead to partial loss of slope information, causing the originally complete slope matrix $\mathcal{G}$ to degrade to $\mathcal{G}^*$, which directly impacts the accuracy of the modal coefficient matrix $\mathcal{A}$ calculations, introducing greater complexity and uncertainty.

$$G^* = \begin{bmatrix} G_x(1) \\ G_y(1) \\ G_x(2) \\ G_y(2) \\ \dots \\ G_x(m^*) \\ G_y(m^*) \end{bmatrix}_{2m^* \times 1} \quad m^* < m \quad (7)$$

Figure 2 displays the residual wavefront reconstructed using modal method for a wavefront significantly affected by strong turbulence. The residual root mean square (RMS) is approximately 1.8275λ , markedly higher compared to normal conditions. When near-field light intensity

fluctuations are strong, leading to under-sampling effects on wavefront detection, this can introduce substantial errors in wavefront reconstruction, failing to meet the fundamental requirements of AO systems.

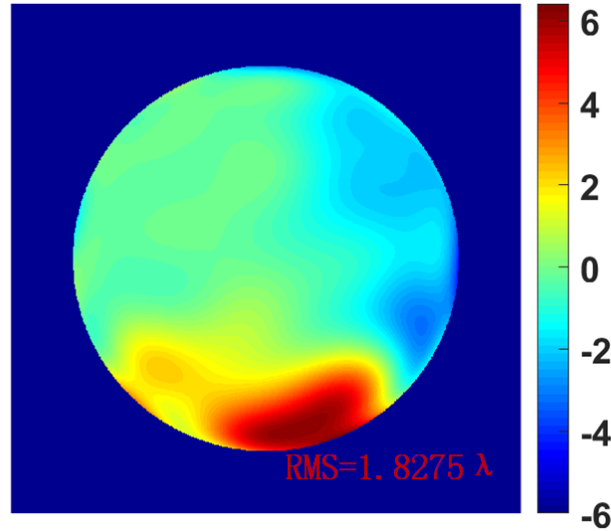


Fig. 2. Residual wavefront reconstructed by conventional methods under conditions of Under-Sampling.

During the detection, the SHWFS inherently fails to capture all nonlinear information of the wavefront. This limitation is further exacerbated in conditions of under-sampling, where traditional threshold processing algorithms are no longer capable of effectively mitigating the impact of intensity fluctuations on slope calculations.

To address this issue, wavefront reconstruction algorithms must adopt additional measures to compensate for the missing information. Traditional methods struggle to maintain both high precision and efficiency in the compensation process. Currently, researches based on neural networks often suffer from a lack of interpretability, insufficient optimization directions, and limited practicality. Therefore, there is a pressing need to delve deeper into neural networks, a potent tool for solving nonlinear problems, to develop new and effective solutions.

3. Mathematical principles of neural network nonlinear activation for under-sampled wavefront slope information

Neural networks face challenges such as insufficient interpretability and a lack of clear optimization directions when processing under-sampled wavefront information, which limits their capability to solve practical problems. Thus, exploring the mathematical principles is crucial for enhancing the precision of wavefront reconstruction. Based on this, this section conducted a detailed analysis of the computation of under-sampled slopes input into the basic modules of convolutional neural networks (CNNs). At the same time, the phenomenon of nonlinear activation of under-sampled wavefront slope elements by CNNs was identified, and a mathematical relationship between the under-sampling ratio and the internal structural parameters of the neural network was established.

3.1. Analysis of under-sampled wavefront processed by a single neural network module

Although CNNs become increasingly complex with additional layers, they are composed of structurally similar basic modules. The repeatability of their structure and the consistency in

their mathematical representation make it possible to analyze the computation of undersampled slope data within a single module, and generalize the mathematical logic in the whole CNN.

The basic module of the CNN is shown in Fig. 3. The input is a one-dimensional slope vector collected by the SHWFS. Convolution kernels of sizes 3 and 5 provide a balance of parameters and computational efficiency for most CNN experimental designs. A stride of 1 preserves more spatial information, captures subtle features, and maximizes learning and feature representation capabilities. Therefore, we initially set convolution kernels to sizes 3 and 5, with a stride of 1, and assign bias values b_1 and b_2 to enhance network flexibility and adjust activation thresholds.

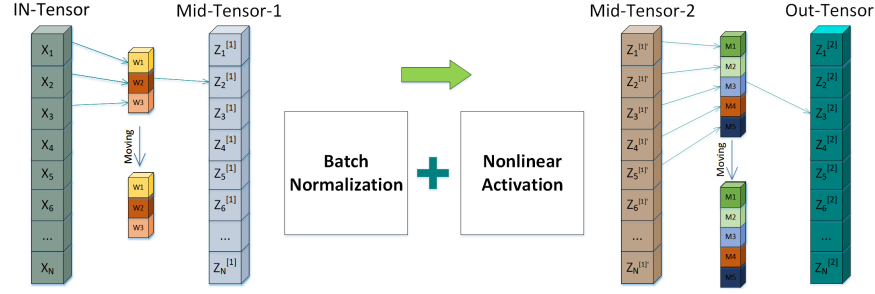


Fig. 3. Schematic of the working principles of a basic neural network module

Given the input slope vector denoted as $\mathcal{X} \in \mathcal{R}^{1 \times N}$, the representation after the first layer of convolution can be expressed as follows:

Herein, $\mathcal{Z}^{[1]}$ denotes the feature vector generated through the first layer of convolution operation applied to the input slope vector, where $\mathcal{W}$ represents the weight parameters of the convolution kernel, and $\mathcal{K}$ signifies the size of the kernel, with $\mathcal{C}$ indicating the number of convolution channels. This operation describes the process of generating new feature vectors through linear combination of any wavefront slope data after the first convolution calculation.

Undersampling results in the input slopes containing multiple zero elements, which can be reused for wavefront reconstruction after computation by the basic CNN module. Figure 4 elucidates the padding process of a single zero element: via convolution kernel operations, surrounding effective elements are fully utilized, and in conjunction with the bias information, new slopes are formed through linear combinations. This step effectively recovers zero elements introduced by undersampling, providing a precise fit foundation for subsequent non-linear activation.

$$\begin{aligned}
 z_1^{[1]} &= 0 \cdot w_1 + x_1 \cdot w_2 + x_2 \cdot w_3 + b_1^{[1]} \\
 z_2^{[1]} &= x_1 \cdot w_1 + x_2 \cdot w_2 + x_3 \cdot w_3 + b_1^{[1]} \\
 &\dots \\
 z_{N-1}^{[1]} &= x_{N-2} \cdot w_1 + x_{N-1} \cdot w_2 + x_N \cdot w_3 + b_1^{[1]} \\
 z_N^{[1]} &= x_{N-1} \cdot w_1 + x_N \cdot w_2 + 0 \cdot w_3 + b_1^{[1]}
 \end{aligned} \tag{8}$$

The aforementioned computation can be summarized in the following equation:

$$z_i^{[1]} = \sum_{k=1}^3 x_{i+k-2} \cdot w_{c,k}^{[1]} + b_c^{[1]}, i = 1, 2, \dots, N; c = 1, 2, \dots, C_1 \tag{9}$$

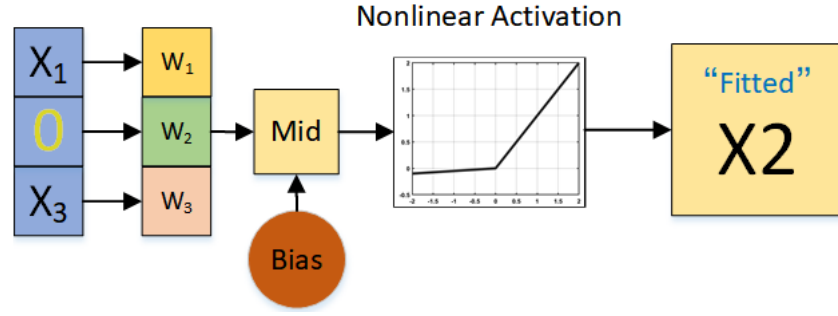


Fig. 4. The “non-linear activation” process for missing slopes. The initial “0” is transformed into a mid-level feature element through a multi-weight linear combination. The feature element is then “activated” by a nonlinear activation function, resulting in a new element X_2 .

Due to the fundamentally nonlinear nature of wavefront under undersampling conditions, the reconstruction process of missing slopes relies solely on a linear combination of adjacent valid elements, which fails to accurately represent the wavefront information at the missing locations. Therefore, the introduction of a nonlinear activation function becomes particularly crucial. This function provides a deep nonlinear characterization for the new slopes generated by the linear combination, corresponding to the nonlinear characteristics of atmospheric turbulence, allowing the generated slopes to be further “activated”. The inclusion of nonlinear activation is the key to the capability of CNN models to reconstruct undersampled wavefronts.

The nonlinear activation process is defined by the function ℓ , in which $\mathcal{P}$ represents the variable parameter within the activation function.

$$f(z) = \begin{cases} z, & \text{if } z \geq 0 \\ p \cdot z, & \text{otherwise} \end{cases} \quad (10)$$

In summary, the mathematical process for generating new elements for any missing position, which undergoes convolution, nonlinear activation, regularization, and a subsequent convolution, is as follows:

$$Z_i^{[2]} = G \left[\sum_{k=1}^5 f(z_{i+k-2}^{[1]}) \cdot W_{c,k}^{[2]} + b_c^{[2]} \right], \quad i = 1, 2, \dots, N; c = 1, 2, \dots, C_2 \quad (11)$$

$$G(z) = \frac{z - \mu}{\sqrt{\delta^2 + \varepsilon}} \quad (12)$$

$$BN = \gamma \cdot G(z) + \beta \quad (13)$$

In the convolution operations, $\mathcal{G}$ denotes the batch normalization process, where μ and δ^2 are the mean and variance of the batch, respectively. The term ε is a small constant introduced to prevent division by zero, while γ and β are the scaling and shifting parameters obtained through learning.

Thus, within the basic module of the CNN, the feature vectors between convolutional layers exhibit the following sequence of operations:

$$\mathbf{Z}^{[n+1]} = \cup_C G \left[\sum_{k=1}^{K_{n+1}} f(z_{i+k-2}^{[n]}) \cdot W_{c,k}^{[n+1]} + b_c^{[n+1]} \right], \quad i = 1, 2, \dots, N; c = 1, 2, \dots, C_n \quad (14)$$

To this point, we analyze the internal computation of undersampled slope data within a single CNN module: The initial fitting convolutional layer captures the intrinsic relationships between

missing slopes and their adjacent slopes, reconstructing the originally zero elements for the first time through a linear combination, as described by Eq. (9). Subsequently, the new elements generated by the linear combination are fed into two convolutional layers, where they encounter embedded nonlinear activation functions, as described by Eq. (11) and Eq. (14), activating their corresponding nonlinear characteristics of atmospheric turbulence. Further, through the action of the final convolutional layer, these characteristics are expanded, thus enriching the expression of features.

Based on the above equations, the input-output relationship of a single CNN module is denoted as $\mathcal{F}_0$, as represented by Eq. (15). Here, 0_i represents a single missing slope, and $\mathcal{Z}_i$ denotes the “activated” feature element after passing through a CNN module. This efficiently describes the functional relationship between the newly “activated” slopes and the input undersampled elements as calculated by the CNN module.

$$\mathcal{Z}_i = \mathcal{F}_0 \left\{ x_{i - \frac{K_1 + K_2 - 2}{2}}, \dots, 0_i, \dots, x_{i + \frac{K_1 + K_2 - 2}{2}} \right\} \quad (15)$$

In a model comprising N sub-modules, assume that the input undersampled zero element is represented as I_i , and the corresponding “activated” output at respective positions is Out_i . The dimensions of each convolutional layer are denoted as $K_1, K_2, \dots, K_{2N}$, and the stride settings are $S_1, S_2, \dots, S_{2N}$. Extending the computational process from the single basic module previously discussed, the functional relationship F between a single output element and input elements within the network model can be derived as follows:

$$\text{Out}_i = F \left\{ x_{i - \frac{K_1 + \sum_{j=1}^{2N-1} \left[(K_{i+1} - 1) \cdot \prod_{j=1}^i S_j \right] - 1}{2}}, \dots, I_i, \dots, x_{i + \frac{K_1 + \sum_{j=1}^{2N-1} \left[(K_{i+1} - 1) \cdot \prod_{j=1}^i S_j \right] - 1}{2}} \right\} \quad (16)$$

According to the above equation, each output element Out_i is formed by “activating” the corresponding undersampled zero input element I_i with its adjacent initial input slopes through a combination of multiple linear combinations and multiple nonlinear activations. In this process, the fundamental modules of the CNN effectively leverage long-range input elements to reconstruct individual missing ones, achieving data completion.

This involves specific configurations of convolutional layer dimensions and strides, as well as the application of nonlinear activation functions, which collectively enhance the network’s ability to “activate” and fill in missing slopes in the input information. As illustrated in Fig. 5, multiple CNN basic modules with varying parameters are employed to iteratively fit and supplement these missing slopes. Each module targets different aspects of the incomplete data, and their combined outputs progressively reconstruct the missing information.

This capability is crucial for reconstructing undersampled wavefronts of varying degrees, as the increase in the undersampling ratio necessitates an enhanced ability of the neural network to fit individual elements. A CNN comprising N fundamental modules “activates” a single output element, with the number of initial input elements involved in the computation being:

$$N = K_1 + \sum_{i=1}^{2N-1} \left[(K_{i+1} - 1) \cdot \prod_{j=1}^i S_j \right] \quad (17)$$

For any ratio of undersampled wavefront, missing slopes can be “activated” through this process. The ultimate activating capability of the CNN model is defined as the proportion of the number of initial input elements corresponding to a single output element within the total initial

elements of the input wavefront—that is, the undersampling ratio of the input wavefront needs to be within this range to ensure effective “activating” of the missing slopes.

Further, theoretically, a model trained with a high undersampling ratio already possesses the capability to fit data under severely limited information conditions. Therefore, within a specific range, models trained with higher undersampling ratio data are equally applicable to normal data, which contains richer effective information. This can be represented as Eq. (18)., for any undersampling ratio r less than r_0 and its corresponding sample subset S_r , the “activating” capability of a model trained at ratio r_0 will not be inferior to that at ratio r .

$$\forall r < r_0, \exists S_r \subseteq S : F(r_0, S) \geq F(r, S_r), r \in [r_0 - r_i, r_0] \quad (18)$$

Notably, this generalization ability primarily relies on the mathematical relationship between the network’s structural design and the undersampling ratio, rather than the specific features of the training data, which can greatly simplify the processing for full-aperture aberration reconstruction of undersampled wavefronts.

To this end, the mathematical relationship between the ratio of nonlinear activation of input slopes and the wavefront undersampling ratio is represented as: (where NumSubspots is the number of subapertures in the SHWFS).

$$\text{UnderSampling}\% < = \frac{K_1 + \sum_{i=1}^{2N-1} \left[(K_{i+1} - 1) \cdot \prod_{j=1}^i S_j \right]}{\text{NumSubspots}} * 100\% \quad (19)$$

This relationship delineates the intrinsic connection between the internal structural parameters of the CNN and its ability to process undersampled wavefront, providing guidance for designing neural networks to reconstruct wavefronts undersampled at different ratios. This insight is immensely valuable for advancing the practical application of such methodologies.

3.2. Validation of the mathematical principles

To verify the proposed theory, a cascaded architecture comprising 30 basic CNN modules was designed in this section, as shown in Fig. 6.

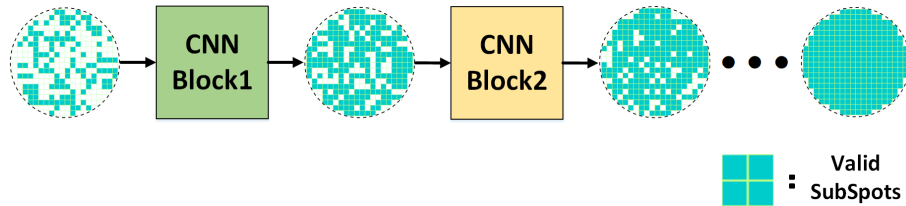


Fig. 5. Schematic of the missing slopes completion process. Multiple CNN basic modules, each with different parameters, iteratively fit and supplement the missing slopes.

After completing the network design, we crafted a dataset for training and testing the network. Specifically, we simulated 16,000 sets of incident wavefronts through numerical simulations. Each wavefront set was constructed from a random linear combination of 65-order Zernike polynomials, excluding the piston term.

These Zernike coefficients were randomly generated based on the Kolmogorov turbulence model, ensuring that the simulation data conformed to the statistical characteristics of atmospheric turbulence. The turbulence strength index, $\mathcal{D}/r_0$,—where $\mathcal{D}$ represents the aperture diameter of the optical system and r_0 represents the atmospheric coherence length—was set within the range of 1 to 32, reflecting the transition from weak to strong turbulence conditions. According

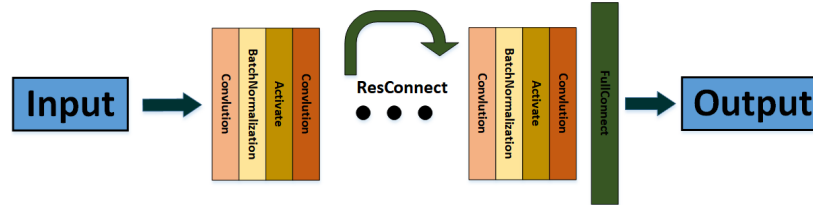


Fig. 6. CNN structure composed of basic modules.

to Eq. (20), the mean square wavefront error σ^2 is given, defining the aperture size over which the mean square wavefront error is 1 rad^2 in phase.

$$\sigma_{\text{Atmosphere}}^2 = 1.0299(D/r_0)^{5/3} \quad (20)$$

However, due to the random statistical nature of atmospheric turbulence and its complex effects on light beams, it is challenging to precisely control and produce slope matrices with specific undersampling ratios. Therefore, a masking strategy was adopted to simulate the random wavefronts at various undersampling ratios, as shown in Fig. 7. This method randomly removed a specific percentage of subapertures, simulating the local loss of slope data caused by near-field light intensity fluctuations.

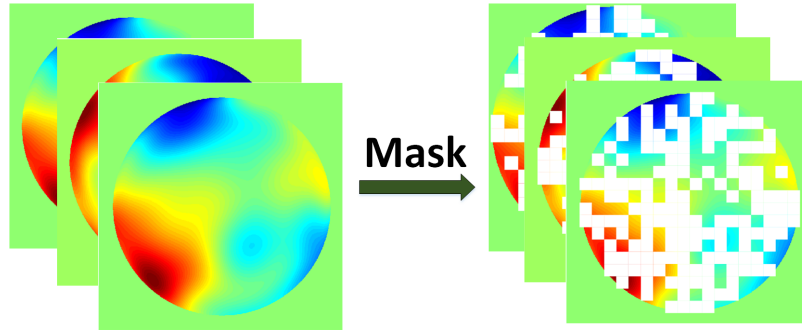


Fig. 7. Schematic of masking for generating undersampled training data. A mask simulates a wavefront affected by turbulence by randomly removing sub-aperture slopes in proportion.

The mathematical process of the neural network “activating” missing slope data, as previously discussed, demonstrates the potential for strong generalization in models trained with data at higher undersampling ratios. To fully explore this potential, the undersampling ratio in the training dataset is uniformly set to 60%. This ratio ensures a high degree of data loss, sufficient to simulate the more severe undersampling conditions that might be encountered in practical applications.

$$\text{Loss} = \sum_{i=1}^M |P_i - R_i| \quad (21)$$

The training process can be simplified into an end-to-end regression model, where the input is the undersampled slope matrix detected by SHWFS and processed with a mask, and the output is the corresponding 65-order Zernike coefficients. Therefore, the robust L1 norm is used as the loss function. In this function, P_i and R_i represent the predicted and actual Zernike coefficients at the i_{th} position, respectively, with M being 65, the order of Zernike coefficients.

Based on Eq. (19), the network parameters were adjusted to accommodate a 60% undersampling ratio, and after multiple adjustments of the initial fitting module and feature mapping module

parameters, we conducted several training sessions. The conclusion was drawn by observing the prediction results of the network model on the same batch test data.

The preliminary feature enhancement model increased the convolution kernel size and stride of the initial fitting module. Compared to the original model, the new one exhibited varying wavefront reconstruction accuracy across different undersampling ratios. As shown in Fig. 8, when the undersampling ratio increased from 0% to 60%, the new model's reconstruction residual RMS is higher than the original model, peaking at 0.262λ (an increase of 0.115λ). However, from 60% to 90%, the new model's residual RMS is lower than the original, with a maximum decrease of 0.161λ to 0.754λ .

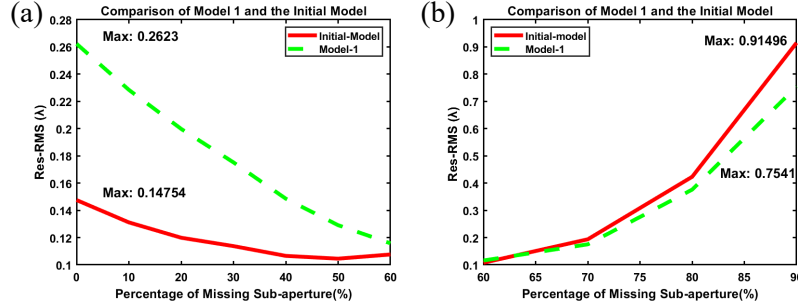


Fig. 8. Comparison of prediction accuracy between the preliminary feature enhancement model and the initial model. (a) Residual RMS between two models across an undersampling ratio range from 0% to 60%. (b) Residual RMS between two models across an undersampling ratio range from 60% to 90%.

Thus, increasing the parameters of the initial fitting module enhances the model's predictive performance for high undersampling rates, but reduces its adaptability to wavefronts at lower undersampling rates.

The feature mapping enhancement model increased the convolution kernel size and stride of the feature integration mapping module. As shown in Fig. 9, the trend in residual RMS changes with the undersampling ratio is consistent with the previous model. Under full light, the maximum residual RMS for this model is 0.202λ , an increase of 0.0543λ compared to the original model but a decrease of 0.06λ compared to the preliminary feature enhancement model. At a 90% undersampling ratio, the maximum residual RMS is 0.714λ , which is 0.201λ lower than the original model and 0.04λ better than the preliminary feature enhancement model.

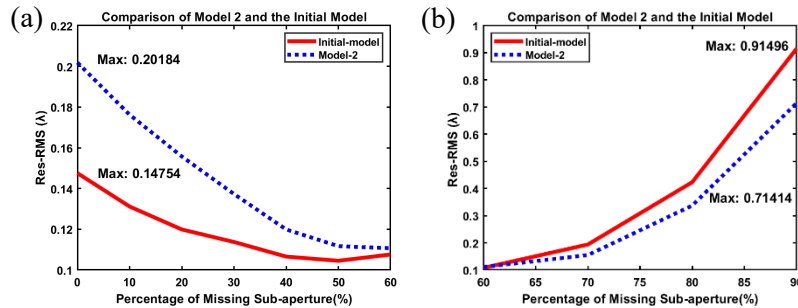


Fig. 9. Comparison of prediction accuracy between the feature mapping enhancement model and the initial model. (a) Residual RMS between two models across an undersampling ratio range from 0% to 60%. (b) Residual RMS between two models across an undersampling ratio range from 60% to 90%.

Based on this, expanding the parameters of the feature integration mapping module improves the performance under high undersampling conditions, surpassing the preliminary feature enhancement model. Moreover, the accuracy loss at full light is also reduced.

The holistic feature enhancement model combined the parameter adjustment strategies of the previous two models, enlarging relevant parameters in both the initial fitting and feature mapping modules. Performance comparisons show that it integrates the characteristics of the aforementioned models. As illustrated in Fig. 10, under full light, its residual RMS is 0.235λ , between 0.202λ and 0.262λ . At a 90% undersampling ratio, the residual RMS is reduced to 0.65λ , demonstrating exceptional performance under extreme undersampling conditions.

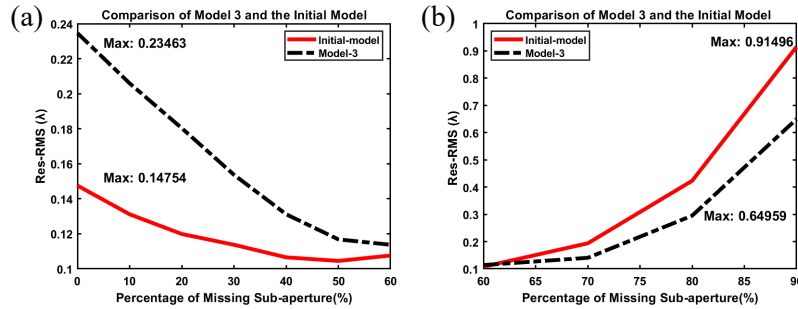


Fig. 10. Comparison of prediction accuracy between the holistic feature enhancement model and the initial model. (a) Residual RMS between the two models across an undersampling ratio range from 0% to 60%. (b) Residual RMS between the two models across an undersampling ratio range from 60% to 90%.

Figure 11(a) illustrates the wavefront reconstruction accuracy across four types of models on the same test set, as the undersampling ratio changes, further verifying the theoretical analysis: maintaining the same number of network layers, using larger convolution kernel sizes and stride lengths can effectively capture long-distance spatial relationships, effectively “activating” missing slopes, and significantly enhancing reconstruction accuracy under high undersampling ratios.

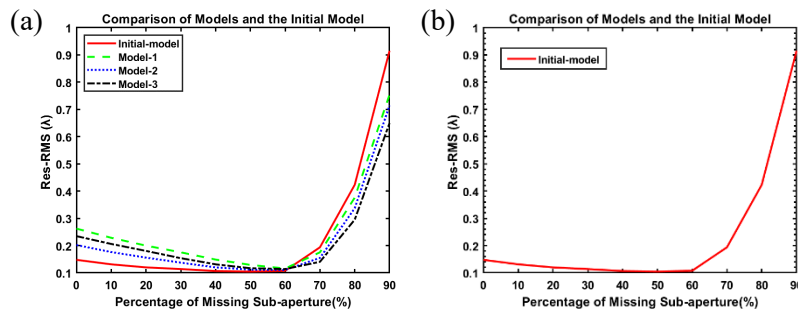


Fig. 11. Comparison of the prediction accuracy across four model types, along with the wavefront residual RMS curve as restored by the initial CNN model.

This analysis procedure extensively discusses the limits of CNN capabilities in fitting undersampled wavefronts. In conjunction with Eq. (19), by increasing the convolution kernel parameters of a single specific module from 3 and 5 to 7 and 11, and the stride from 1 to 2, the theoretical ratio for processing undersampled wavefronts increases to approximately 63%, and the network model’s prediction accuracy for test data with a 90% undersampling ratio improves by 17.69% and 21.97%. By simultaneously enlarging the parameters of both modules, the prediction accuracy for test data with a 90% undersampling ratio improves by approximately 25.68%. This

process indicates that the fitting capability of a neural network improves as it relies on a larger proportion of input slopes to “activate” a single missing slope.

Despite the theoretical advantage that models trained with higher undersampling ratio data have stronger generalization capabilities for data with richer effective information, as shown in Fig. 11, the model performance remains stable within a 10% range as the undersampling ratio decreases, and then continuously declines. This phenomenon can be attributed to the fact that models trained with high undersampling ratio data tend to overuse the available data under low undersampling conditions, resulting in overfitting and limiting the model’s generalization ability to within a 10% ratio range.

Therefore, to prevent a decline in prediction accuracy due to large differences in undersampling ratios, it is crucial to divide the ratio interval into 10% segments during subsequent training processes. By providing the highest ratio of undersampled data within each interval to the model, this strategy ensures that the model maintains good generalization performance across the entire range of undersampling ratios.

4. Network model improvements and simulation validation

In section 3, a CNN model was built to validate our proposed theory. However, due to the relatively simple structure of the validation model, to meet the practical applications of AO systems, it is inevitably necessary to increase the depth of the network. While this approach enhances the model’s capabilities, it also significantly increases computational complexity, severely reducing the network’s convergence speed and affecting training efficiency.

To address this challenge, we considered incorporating the widely-used Self-Attention mechanism in the process of network improvement, which is commonly applied in sequence processing tasks [35]. The mathematical principles of this mechanism can be referenced in Eq. (22–24). Specifically, the “query” represents the self-representation of the target element; the role of the “key” is to match with the “query” to determine the relative importance of each element; the “value” calculates the actual contribution of the elements based on the determined importance. Additionally, W represents a learnable weight matrix, while the softmax function normalizes the contribution scores, ensuring that the total contribution of all elements sums to one.

$$q_i = \sum_{j=1}^i x_{i+j-1} \cdot W_j^q; k_i = \sum_{j=1}^i x_{i+j-1} \cdot W_j^k; v_i = \sum_{j=1}^i x_{i+j-1} \cdot W_j^v \quad (22)$$

$$Softmax = \frac{e^x W_j}{\sum_{k=1}^K e^x W_k} \quad (23)$$

$$Attention(q, k, v) = Softmax\left(\frac{q \cdot k^T}{\sqrt{d_k}}\right) \cdot v \quad (24)$$

This mechanism focuses on the spatial location areas of input information, allowing the model to consider all other elements in the sequence when processing each element, thereby achieving global dependency modeling. This approach is highly congruent with the mathematical logic of neural networks when processing the undersampled wavefronts. Simultaneously, the mechanism also effectively reduces the complexity of the network architecture and conserves computational resources. Its application not only ensures the performance of the model but also significantly enhances its efficiency.

Consequently, we redesigned a deep neural network composed of fully connected layers, convolutional and activation layers, Self-Attention structures, and residual connections for the reconstruction of undersampled wavefronts. As illustrated in Fig. 12, the network comprises seven sub-modules, each consisting of convolution, regularization, PReLU nonlinear activation, and Self-Attention structures. Apart from differences in convolution kernel sizes and channel

numbers, the modules are similar; we refer to it as the Undersampled Wavefront Reconstruction Net model (UWR-Net). Since the undersampled slopes are detected by the SHWFS, to further minimize information loss, we omitted the use of pooling and encoder-decoder structures in our design.

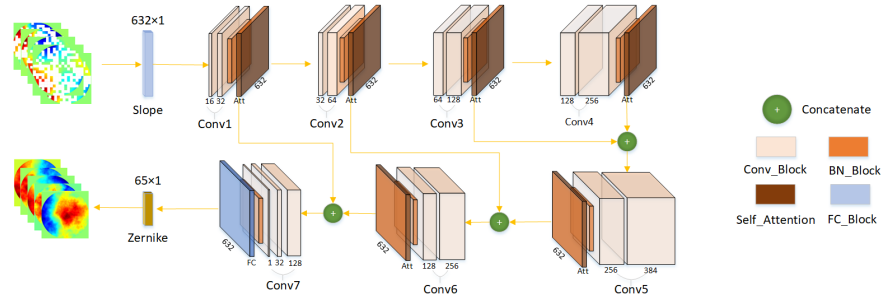


Fig. 12. Schematic of the UWR-Net architecture. Pink layers represent convolution kernels with varying parameters, orange layers denote batch normalization modules, brown layers signify the self-attention mechanism, and blue layers correspond to fully connected layers. Green circles indicate residual connections.

Initially, the undersampled slopes are expanded in four dimensions. The initial fitting module consists of a 3×1 convolutional layer followed by a 5×1 convolutional layer, connected by a batch normalization (BN) and a PReLU activation layer. The convolution kernel sizes for the second to fourth sub-modules are set as 3×1 , 3×1 , 5×1 , 5×1 , and 7×1 , 7×1 , respectively. Different sizes are used for the one-dimensional convolutions in various sub-modules to ensure

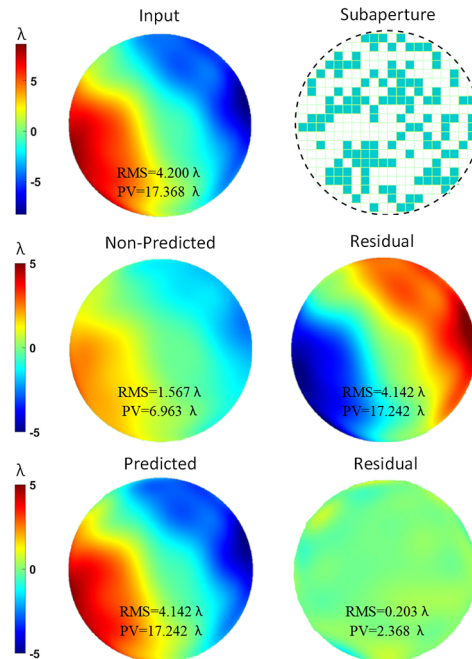


Fig. 13. Visualization of reconstruction effects on a 60% undersampled wavefront. “Non-predicted” refers to the modal method, while “Predicted” refers to the proposed method. The residual is the difference between the reconstructed and input wavefronts.

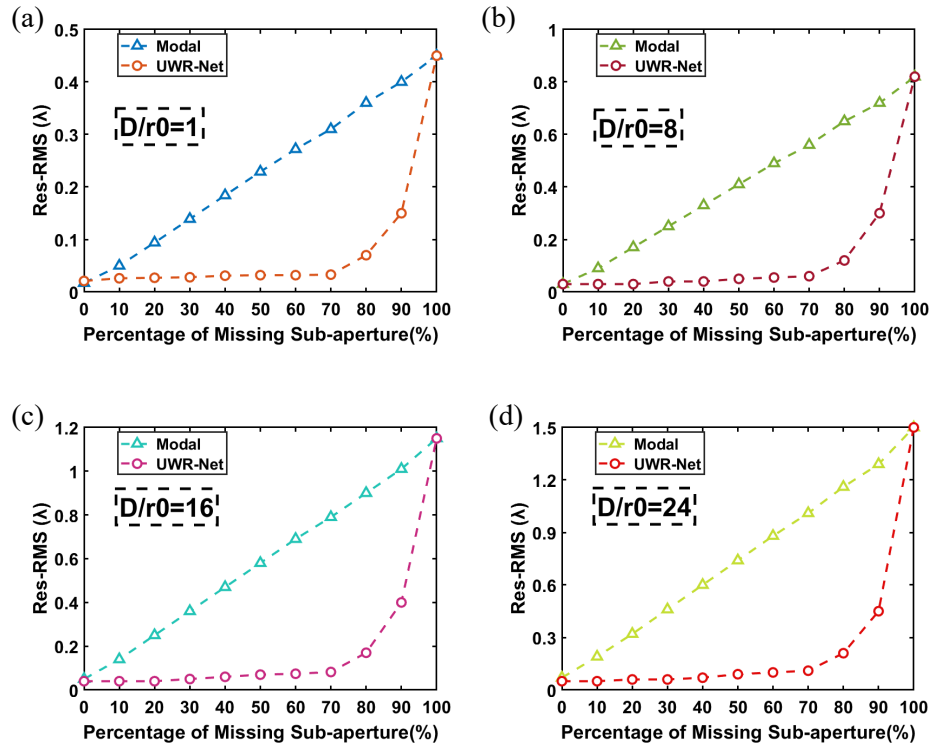


Fig. 14. The comparison of the UWR-Net model's reconstruction results for undersampled wavefronts subjected to four different intensities of turbulence. (a) $D/r_0 = 1$. (b) $D/r_0 = 8$. (c) $D/r_0 = 16$. (d) $D/r_0 = 24$.

that they can extract multi-level feature information with different receptive fields. The expanded multi-channel features are then integrated by a dimension reduction sub-module to reconstruct the initial data's channel dimension. The reduction process consists of a similar three-layer structure. Finally, the corresponding 65th-order Zernike coefficients matrix is output through a 1×1 convolutional layer and a fully connected layer.

This model used the same training parameter as the previous validation process. To avoid overfitting, its training set was expanded from a single undersampling ratio to multiple ratios. Specifically, the entire range was divided into steps of 10%, providing the model with 0% to 70% undersampled wavefront data for training.

Figure 13 displays the model's prediction results for a frame of 60% undersampled wavefront. The results show that the randomly input wavefront has an RMS value of 4.2λ . The modal method, due to the effects of undersampling, struggles to effectively reconstruct the initial wavefront, with a residual RMS value of 4.14λ . In contrast, our method is more effective, reducing the residual RMS to only 0.203λ , a decrease of approximately 95%. This result convincingly demonstrates the significant advantage of this approach in addressing high undersampling ratio wavefront reconstruction challenges.

To rigorously assess the performance of the enhanced network model, a test dataset containing 4000 samples was constructed based on different turbulence parameters $D/r_0 = 1, 8, 16, 24$. These samples were then masked according to undersampling ratios ranging from 0% to 90%. We subsequently compared the reconstruction accuracy using our method against the modal method. This approach allowed us to evaluate the efficacy of the improved model under varying conditions, providing a comprehensive understanding of its capabilities in realistic scenarios.

Figure 14 illustrates the trends in the mean residual RMS values of the wavefronts reconstructed by two different methods under four turbulence intensities as the undersampling ratio increases. Specifically, as the undersampling ratio increases, the residual RMS of the wavefronts reconstructed by the modal method shows a linear increasing trend, revealing the limitations of the modal method in addressing highly undersampled wavefronts, especially under strong turbulence conditions. In contrast, the mean residual RMS of the wavefronts reconstructed by the proposed method remains relatively stable when the undersampling ratio is $\leq 70\%$, reducing the residual RMS by up to 89.3% compared to the modal method.

These results indicate that the improved model not only simplifies the network structure but also demonstrates significant advantages over the modal method in dealing with full-aperture aberration reconstruction for undersampled wavefronts, showing high accuracy and generalizability across different levels of undersampling. These findings not only revalidate the accuracy of the theoretical analysis but also provide a solid foundation for the practical deployment of this model in applications involving the reconstruction of undersampled wavefronts.

5. Experiment

In this section, the effectiveness of the proposed method is validated through experiments. The experimental setup, as depicted in Fig. 15 and Fig. 16, begins with a laser source emitting a beam at a wavelength of 976 nm. Initially, this beam is shaped by an aperture diaphragm (AP) to obtain a clear spatial profile. Subsequently, the laser intensity is adjusted through an attenuator (AT). The preliminarily shaped beam then passes through a turbulence generator, which uses alternating hot and cold air flows to simulate atmospheric disturbances, thereby introducing various levels of wavefront aberrations into the beam. Table 1 shows the key parameters of the experimental system.

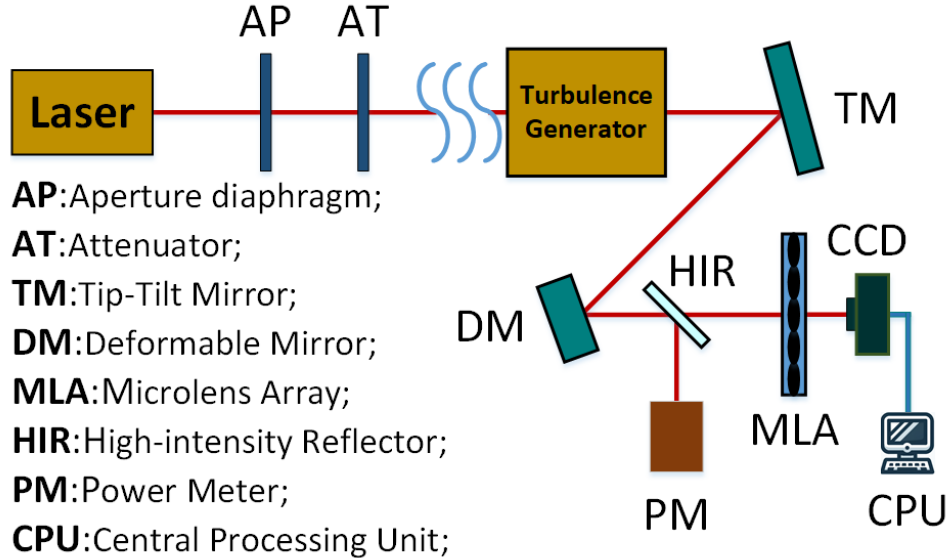


Fig. 15. Schematic of the optical path.

The beam, affected by turbulence, is reflected by a series of optical components, including a tilt mirror (TM), a deformable mirror (DM), and a high-intensity beam splitter (HIR), ultimately aligning with the SHWFS. This sensor is composed of a micro-lens array (MLA) and a charge-coupled device (CCD) camera. The micro-lens array (MLA) plays a pivotal role in the wavefront

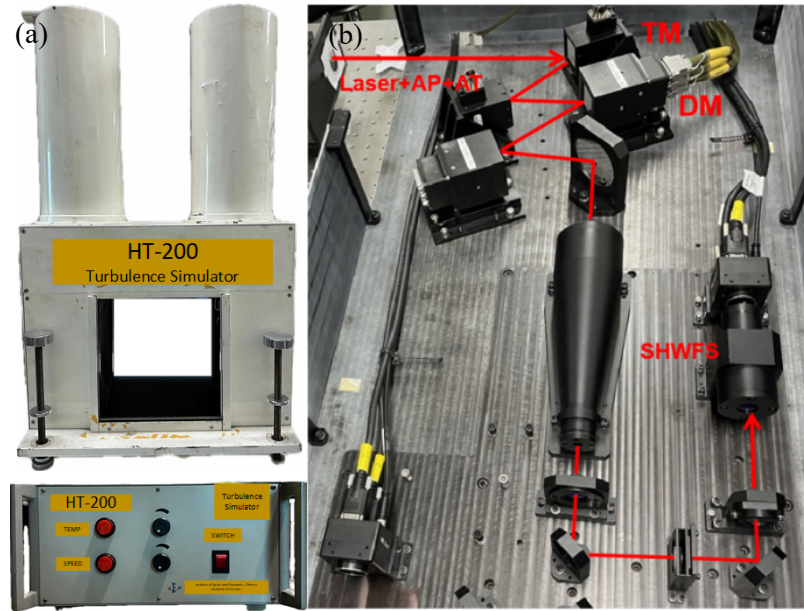


Fig. 16. The HT200 turbulence simulator and its control apparatus, as well as the experimental platform optical path. (a) the HT200 turbulence simulator, (b) the experiment platform.

Table 1. Key Parameters of the Experiment.

Parameters	Values
wavelength	976nm
microlens Size	352um
microlenses numbers	20 × 20
focal length of the microlens	16mm
valid sub-apertures numbers	316
pixel size	11um × 11um

sampling process, allowing the CCD camera to precisely capture the image of the spot array, which provides fundamental data for subsequent slope matrix calculations and wavefront reconstruction.

Figure 16(a) depicts the HT-200 type turbulence generator used in the actual experimental system. This device consists of an air convection unit and a temperature and wind speed control unit, capable of generating airflows with specific wind speeds and temperature gradients.

The temperature control unit in the turbulence generator provides stable heat output, while the wind speed control and convection devices are responsible for producing a uniform airflow distribution, ensuring that the generated atmospheric disturbances match real atmospheric turbulence. This precise simulation creates an ideal experimental platform for introducing and analyzing wavefront aberrations.

We utilized this platform to collect multiple sets of wavefront data to validate the applicability and robustness of the proposed method. To generate the random wavefront disturbances needed for testing, the turbulence simulator was controlled to produce atmospheric turbulence with D/r_0 approximately ranging from 1 to 12. The SHWFS subsequently collected 600 sets of random wavefronts and used the centroid method to calculate their slope information. As shown in Fig. 17, the random incident wavefronts captured by the SHWFS camera in the system is presented.

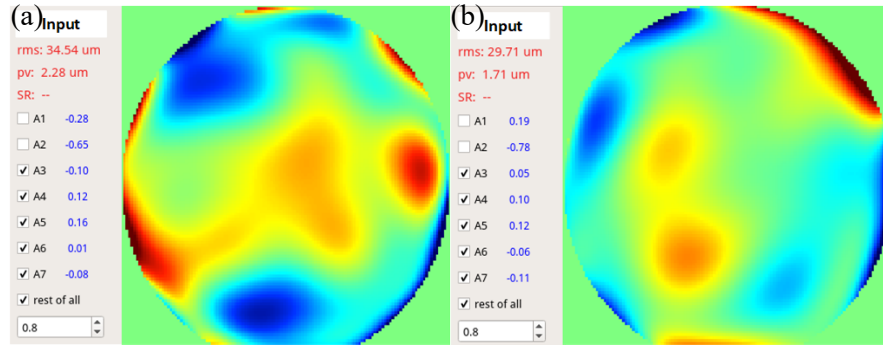


Fig. 17. The typical wavefronts affected by generated turbulence.

By masking the obtained incident wavefront to simulate a 0%-90% slope information loss, the undersampled wavefronts were formed. These slope matrices were then input into the model UWR-Net to predict the corresponding Zernike coefficients and reconstruct the wavefront. Likewise, the proposed method was compared with other methods.

Figure 18 displays the comparison of the mean residual RMS of wavefront reconstruction between the modal method and our research method under 30%, 50%, and 70% undersampling conditions, using $1/15\lambda$ as a benchmark for measuring reconstruction effectiveness. It is observable that the modal method only rarely approaches the benchmark level under any undersampling condition. In contrast, the mean residual RMS predicted by the UWR-Net is predominantly well below the benchmark, with a reduction of up to 90% compared to the modal method, demonstrating significant superior performance.

Furthermore, Fig. 19 presents the residual wavefront reconstructed from 70% undersampled wavefront in a 3D format. We compare the residual wavefronts reconstructed by the modal method and the proposed method. The top four panels show the residual wavefronts obtained using the modal method, which exhibit poor uniformity and noticeable fluctuations, with a peak-to-valley (PV) value reaching approximately 10λ . In contrast, the lower panels display the residual wavefronts reconstructed by the proposed method, demonstrating significantly improved uniformity with a maximum PV value of around 2λ , representing an 80% reduction compared to the modal method.

To ensure a thorough comparison, we replicated and modified the SH-U-Net model designed by Zhao et al [27]. to reconstruct wavefronts at various undersampling ratios. The replicated SH-U-Net retains a CNN-based internal structure, consisting of 3 downsampling modules and 3 upsampling modules. Excluding skip connections, it comprises approximately 20 convolutional layers—about 6 more layers than our UWR-Net, which was designed based on mathematical derivations.

Additionally, SH-U-Net's maximum number of feature channels extends to 512, which is 128 more than UWR-Net's maximum of 384. From the perspective of parameter count, UWR-Net, with its mathematically optimized design, is more compact and efficient. Figure 20 illustrates the changes in residual RMS as the undersampling ratio increases for the modal method, SH-U-Net, and UWR-Net.

Consistent with the numerical simulation, the residual RMS of the wavefronts reconstructed by the modal method shows a linear increasing trend as the undersampling ratio increases, with reconstruction errors significantly higher than those of both SH-U-Net and UWR-Net. Although SH-U-Net maintains high accuracy at undersampling ratios between 0% and 50%, it still lags behind UWR-Net. Moreover, when the undersampling ratio exceeds 50%, the prediction accuracy of SH-U-Net degrades significantly, whereas UWR-Net continues to perform well.

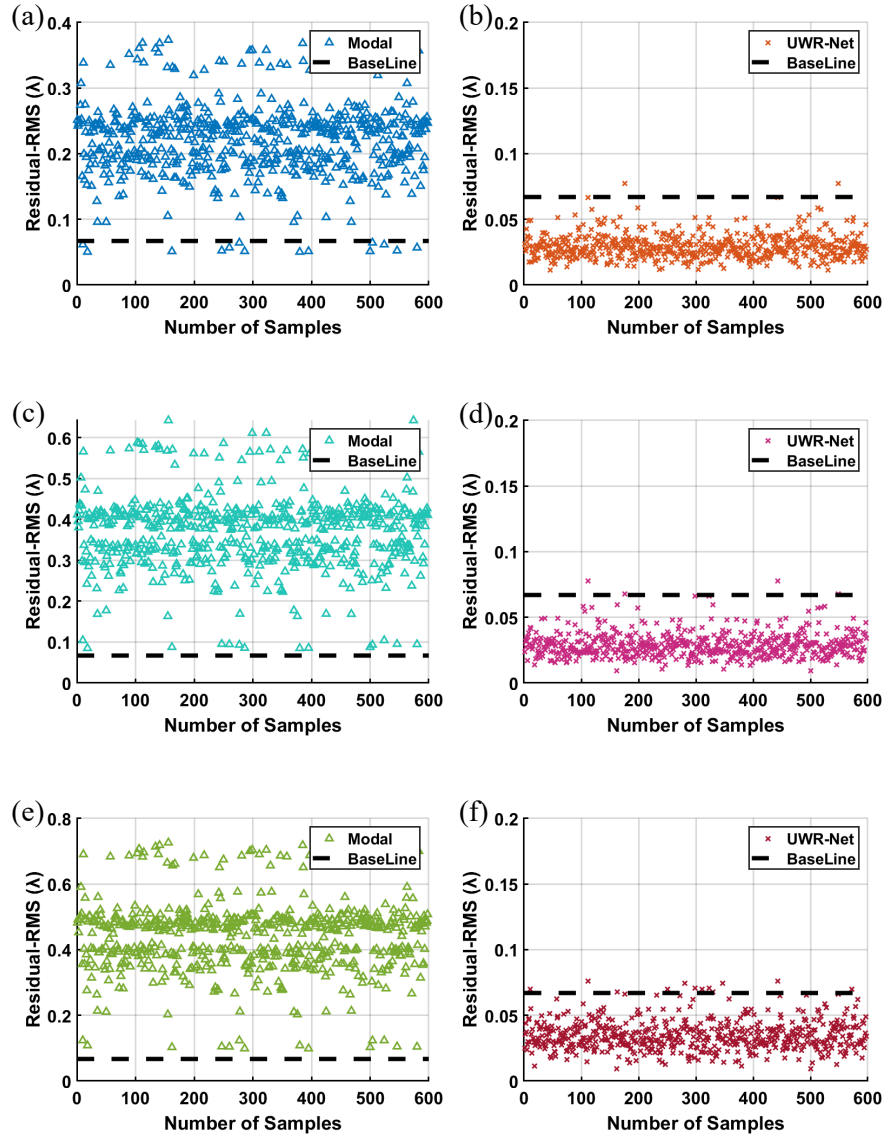


Fig. 18. Comparison of residual RMS for wavefront reconstruction under 30%, 50%, and 70% undersampling conditions using two methods. (a) Modal method at 30% undersampling. (b) UWR-Net at 30% undersampling. (c) Modal method at 50% undersampling. (d) UWR-Net at 50% undersampling. (e) Modal method at 70% undersampling. (f) UWR-Net at 70% undersampling.

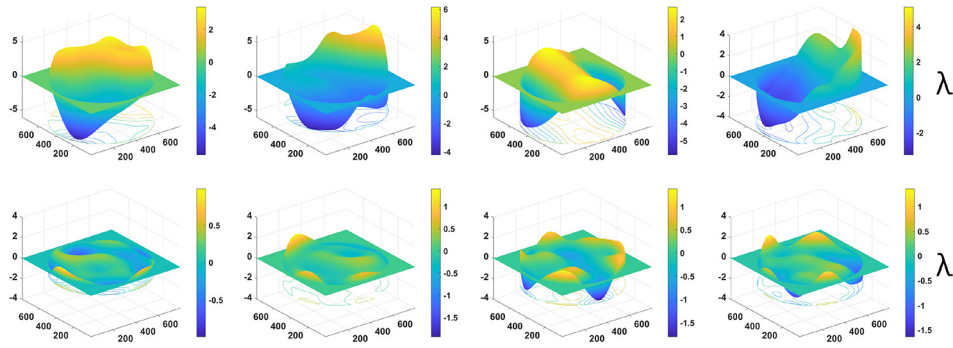


Fig. 19. Comparison of residual RMS for wavefront reconstruction under 70% undersampling conditions using two methods. The top four panels illustrate the residual wavefronts reconstructed using the modal method, whereas the lower panels depict the corresponding residual wavefronts obtained through the proposed method.

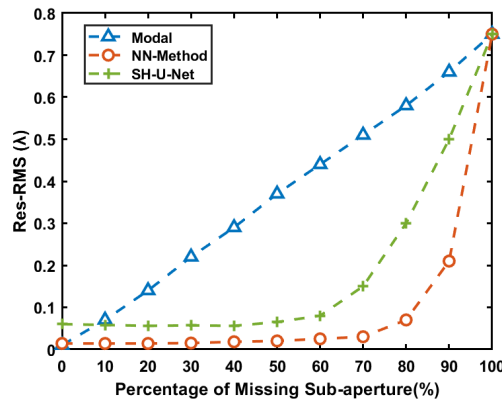


Fig. 20. Comparison of wavefront reconstruction residual RMS between the modal method, SH-U-Net, and UWR-Net in the experiment.

Notably, unlike SH-U-Net, UWR-Net was trained solely on simulation data, yet it achieved benchmark-level wavefront reconstruction.

These experimental results demonstrate that the UWR-Net, designed based on theoretical analysis and trained only with undersampled simulation data, can effectively perform reconstruction of highly undersampled wavefronts in practical applications, exhibiting significant Sim2Real characteristics. This conclusion further emphasizes the important role of theoretical guidance in the design of network models and also showcases the practical application potential of this network structure in tasks involving full-aperture aberration reconstruction at high undersampling ratios.

6. Conclusion

In this study, we investigated the mathematical principles underlying the nonlinear activation of neural networks when processing undersampled wavefront slope information. We proposed and effectively validated a mathematical relationship between the undersampling ratio and the proportion of input slopes activated by the neural network. Both simulation and experimental results demonstrate that UWR-Net, designed based on these mathematical principles, can effectively achieve full-aperture aberration reconstruction even when the wavefront is undersampled by up

to 70%. Compared to the traditional modal method, UWR-Net achieves a maximum reduction in residual wavefront RMS of approximately 89.3%.

The application of neural networks in adaptive optics often faces challenges related to interpretability and practicality. Our work addresses these issues by establishing a mathematical relationship between undersampled wavefronts and neural network parameters, thereby enhancing model interpretability and simplifying training and deployment—including simulation-to-real-world (Sim2Real) transitions.

Given the inherent nonlinearity and complexity of neural networks, our proposed mathematical relationships primarily focus on convolutional kernel parameters and undersampled element ratios. Future research will explore optimizing network architectures, including network channels, activation functions, and attention mechanisms, to further improve performance. Additionally, we aim to mitigate severe information loss caused by obstructions, thereby extending the applicability of our approach to a broader range of adaptive optics applications.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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